

GEOGRAPHY 03 — VOLCANISM AND EARTHQUAKES / INDIA SEISMIC ZONES

ENDOGENIC SYSTEM: PLATE MOTION → EARTHQUAKE SOURCE GEOMETRY, DEPTH → SEISMIC WAVES, SHADOW ZONES → GLOBAL SEISMIC BELTS, CASCADING → VOLCANO ANATOMY, MAGMA CONTROLS → VOLCANIC SETTINGS, GLOBAL BELTS

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure: the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g7 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00 STAGE 00 ENDOGENIC SYSTEM: PLATE MOTION, STRAIN RELEASE AND MAGMA GENERATION

ENDOGENIC SYSTEM

ENDOGENIC SYSTEM

PLATE MOTION, STRAIN RELEASE AND MAGMA GENERATION

INTERNAL HEAT + GRAVITY + PLATE MOTION RESHAPE CRUST.

STRESS ACCUMULATION MAGMA GENERATION LOAD

STRAIN DIVERGENCE

DECOMPRESSION SUBDUCTION

VOLATILE FLUX RUPTURE

SLIP PLUME

ENDOGENIC SYSTEM

Internal heat + gravity + plate motion reshape crust.

STRESS ACCUMULATION MAGMA GENERATION load → lock → strain divergence: decompression subduction: volatile flux rupture

anomalous upwelling

EARTHQUAKE VULCANISM landforms + hazards + resources

DEFINITION / COMPONENTS

ENDOGENIC SYSTEM

Internal heat + gravity + plate motion reshape crust.

STRESS ACCUMULATION MAGMA GENERATION load → lock → strain divergence: decompression subduction: volatile flux rupture → slip plume

TRIGGER / MECHANISM

anomalous upwelling

EARTHQUAKE VULCANISM landforms + hazards + resources

FAULT PAIRS tension → normal

EFFECT / INTERVENTION

compression → reverse/thrust

shear → strike-slip

Plate motion loads a region; sudden fault rupture is the immediate quake source.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Plate tectonics gives the spatial grammar of seismicity and vulcanism, but the examiner still expects the immediate mechanics—fault slip for an earthquake and magma ascent for an eruption.

01 STAGE 01 EARTHQUAKE SOURCE GEOMETRY, DEPTH AND SEISMOGRAM LOGIC

EARTHQUAKE SOURCE GEOMETRY, DEPTH AND SEISMOGRAM LOGIC

SURFACE STATION A ----- EPICENTRE ----- STATION B FOCUS

HYPOCENTRE RUPTURE BEGINS ON FAULT PLANE

SEISMOGRAM ARRIVAL ORDER

P FIRST

S SECOND

LOVE AND RAYLEIGH LATER

P-S INTERVAL AT ONE STATION GIVES SOURCE DISTANCE THREE

SURFACE station A ----- epicentre ----- station B focus / hypocentre rupture begins on fault

SEISMOGRAM ARRIVAL ORDER

P first → S second → Love and Rayleigh later

P-S interval at one station gives source distance three or more distance circles estimate the

SYSTEM / DEFINITION

SURFACE station A ----- epicentre ----- station B focus / hypocentre rupture begins on fault plane

SEISMOGRAM ARRIVAL ORDER

P first → S second → Love and Rayleigh later

PROCESS / MECHANISM

P-S interval at one station gives source distance three or more distance circles estimate the epicentre

DEPTH CLASS shallow: 0-70 km

intermediate: 70-300 km

SPATIAL / INDIA PATTERN

deep: 300-700 km

SEQUENCE FIREWALL foreshock, mainshock and aftershock are relational labels

a swarm has no single clearly dominant event.

HAZARD / LIMIT / RESPONSE

Epicentre is a vertical projection, not necessarily the worst-damaged place.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SEQUENCE FIREWALL foreshock, mainshock and aftershock are relational labels.

02 STAGE 02 SEISMIC WAVES, SHADOW ZONES, MAGNITUDE AND INTENSITY

SEISMIC WAVES, SHADOW ZONES, MAGNITUDE AND INTENSITY

MAIN CLUE

SOLID, LIQUID

CORE REFRACTION

TRANSVERSE SHEAR

SOLID ONLY

LIQUID OUTER CORE

HORIZONTAL SHEAR

WAVE

MOTION

MEDIUM

MAIN CLUE

P

compression

solid, liquid

core refraction

S

transverse shear

solid only

liquid outer core

Love

horizontal shear

surface

lateral shaking

Rayleigh

rolling ellipse

surface

mixed motion

P shadow about 103-142 deg

direct S shadow beyond about 103 deg

DEFINITION / COMPONENTS

MAIN CLUE

solid, liquid

core refraction

transverse shear

solid only

TRIGGER / MECHANISM

liquid outer core

horizontal shear

lateral shaking

rolling ellipse

mixed motion

EFFECT / INTERVENTION

P shadow about 103-142 deg

direct S shadow beyond about 103 deg

MAGNITUDE: one source measure; Mw uses seismic moment moment = rigidity x fault area x average slip

1 local magnitude gives about 10x amplitude and 31.6x energy

INTENSITY: place-specific observed effects; many values for one event source x path x site x exposure x vulnerability / capacity → damage

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Focal depth is a source descriptor, whereas disaster severity is a relational outcome produced by source, path, site, exposure and vulnerability.

03 STAGE 03 GLOBAL SEISMIC BELTS, CASCADING HAZARDS AND TSUNAMI SEQUENCE

GLOBAL SEISMIC BELTS, CASCADING HAZARDS AND TSUNAMI SEQUENCE

GLOBAL EARTHQUAKE DISTRIBUTION

RIDGES/TRANSFORMS TRENCH + SLAB + ARC

HIMALAYAN COLLISION

SHALLOW EVENTS MEGATHRUST + TSUNAMI

COMPLEX CONVERGENCE

BASALTIC RIDGES

INTRAPLATE EVENTS REACTIVATE INHERITED STRUCTURES; STABLE IS NOT ASEISMIC.

GLOBAL EARTHQUAKE DISTRIBUTION

Ridges/transforms trench + slab + arc

Himalayan collision

shallow events megathrust + tsunami

complex convergence

DEFINITION / COMPONENTS

GLOBAL EARTHQUAKE DISTRIBUTION

Ridges/transforms trench + slab + arc

Himalayan collision

shallow events megathrust + tsunami

complex convergence

TRIGGER / MECHANISM

basaltic ridges

Intraplate events reactivate inherited structures; stable is not aseismic.

CASCADE shaking → amplification / liquefaction / lateral spreading

landslide / fire / lifeline failure / dam distress

TSUNAMI submarine megathrust → vertical sea-floor displacement

EFFECT / INTERVENTION

long low deep-ocean wave → coastal shoaling

run-up + inundation + return flow

First wave need not be largest; withdrawal need not occur before every event.

Risk varies with bathymetry, bay shape, shelf, exposure and evacuation.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Depth distribution is the diagnostic map layer: only a descending slab generates a systematic inclined belt from shallow trench earthquakes to intermediate and deep events.

04 STAGE 04 VOLCANO ANATOMY, MAGMA CONTROLS, LANDFORMS AND ERUPTION STYLES

VOLCANO ANATOMY, MAGMA CONTROLS, LANDFORMS AND ERUPTION STYLES

ASH PLUME

CRATER LAVA --- LAVA VENT DYKE

SILL CONDUIT MAGMA CHAMBER

MAGMA BELOW SURFACE

LAVA AFTER ERUPTION

SILICA UP

VISCOSITY UP

crater lava --- lava vent dyke

sill conduit magma chamber

Magma below surface

ash plume

CLASSIFICATION

ash plume

crater lava --- lava vent dyke

sill conduit magma chamber

Magma below surface

DIAGNOSTIC BASIS

lava after eruption

Silica up → viscosity up → degassing harder → explosivity may rise

Temperature up → viscosity generally down

Retained gas + decompression → bubbles → fragmentation

PROCESS LINK

FORMS: batholith, laccolith, lopolith, phacolith, dyke, sill

Dyke cuts layers; sill follows layers.

STYLE: fissure/icelandic → Hawaiian → Strombolian → Vulcanian → Plinian

FORM: shield

EXAM TRAP

cinder cone

fissure plateau

STATUS: active

extinct; status is not a compulsory life cycle.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Active status concerns eruptive capability; shield/composite concerns form; Hawaiian/Plinian concerns behaviour—UPSC can mix these axes in close options.

05 STAGE 05 VOLCANIC SETTINGS, GLOBAL BELTS, PRODUCTS AND ENVIRONMENTAL EFFECTS

VOLCANIC SETTINGS, GLOBAL BELTS, PRODUCTS AND ENVIRONMENTAL EFFECTS

MELTING MECHANISM

RIDGE, RIFT, BASALT

SLAB VOLATILE FLUX

TRENCH-SLAB-ARC SYSTEM

MANTLE UPWELLING

FLOOD BASALT, HOT SPOT

RING OF FIRE

SETTING

MELTING MECHANISM

EXPRESSION

Divergent

decompression

ridge, rift, basalt

Subduction

slab volatile flux

trench-slab-arc system

Plume

mantle upwelling

flood basalt, hot spot

BENEFITS: new crust/land

soil parent

geothermal energy

CLASSIFICATION

MELTING MECHANISM

ridge, rift, basalt

slab volatile flux

trench-slab-arc system

DIAGNOSTIC BASIS

mantle upwelling

flood basalt, hot spot

Ring of Fire = trench + descending slab + earthquakes + volcanic arc

Plume model: broad head may form flood basalt; tail may form age chain.

PROCESS LINK

Deccan Trap is a fissure-fed flood-basalt plateau, not a giant cone.

HAZARDS: lava

density currents

CLIMATE: stratospheric sulphate aerosol can cause temporary cooling

EXAM TRAP

BENEFITS: new crust/land

soil parent

geothermal energy

Impact converges through wind, relief, drainage, exposure and preparedness.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Deccan Trap is a fissure-fed flood-basalt plateau, not a giant cone.

06 STAGE 06 INDIA SEISMIC GEOGRAPHY, ZONATION, INSTITUTIONS AND RESILIENCE STACK

INDIA SEISMIC GEOGRAPHY, ZONATION, INSTITUTIONS AND RESILIENCE STACK

INDIA REGIONAL HIERARCHY

COLLISION, THRUSTS, SLOPES, DENSE CORRIDORS

CONVERGENCE, STRIKE-SLIP COMPLEXITY

SUBDUCTION, TSUNAMI, BARREN ISLAND VOLCANISM

INHERITED RIFT AND FAULT REACTIVATION

RELATIVELY STABLE; KOYNA AND LATUR SHOW RESIDUAL RISK

ALLUVIUM, AMPLIFICATION AND HIGH EXPOSURE

INDIA REGIONAL HIERARCHY

Himalaya : collision, thrusts, slopes, dense corridors

North-East : convergence, strike-slip complexity

Andaman : subduction, tsunami, Barren Island volcanism

SYSTEM / DEFINITION

INDIA REGIONAL HIERARCHY

Himalaya : collision, thrusts, slopes, dense corridors

North-East : convergence, strike-slip complexity

Andaman : subduction, tsunami, Barren Island volcanism

Kachchh : inherited rift and fault reactivation

PROCESS / MECHANISM

Peninsula : relatively stable; Koyna and Latur show residual risk

Indo-Gangetic : alluvium, amplification and high exposure

MACROZONATION: broad design hazard, not prediction

MICROZONATION: geology + soil + motion + water + liquefaction + GIS

NCS/MoSES → monitoring

SPATIAL / INDIA PATTERN

GSI → geology

BIS → standards

NDMA/NIDM → guidance

INCOIS → tsunami warning

States/ULBs → approvals + inspection + enforcement site → regular load path → ductility → connections

HAZARD / LIMIT / RESPONSE

non-structural safety → quality control → retrofit → drills

Early warning detects rupture after it starts; it is not prediction.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Macrozonation tells a city the regional design problem; microzonation tells neighbourhoods how local ground modifies that problem.

07 STAGE 07 HAZARD-RISK-RESPONSE AND MAINS ANSWER SPINE FOR SEISMIC INDIA

HAZARD-RISK-RESPONSE AND MAINS ANSWER SPINE FOR SEISMIC INDIA

HAZARD-RISK-RESPONSE AND MAINS ANSWER SPINE FOR SEISMIC INDIA

WHY CAN SIMILAR EARTHQUAKES PRODUCE UNEQUAL DISASTERS?

SOURCE SIZE + DEPTH + RUPTURE GEOMETRY + PATH

BASIN SEDIMENT + GROUNDWATER + SLOPE + LIQUEFACTION POTENTIAL

PEOPLE + BUILDINGS + LIFELINES + TIME OF OCCURRENCE

WEAK DESIGN + IRREGULAR FORM + POOR MAINTENANCE

CODES + ENFORCEMENT + RETROFIT + WARNING + RESPONSE

RISK VERDICT

QUESTION: Why can similar earthquakes produce unequal disasters?

HAZARD: source size + depth + rupture geometry + path

SITE: basin sediment + groundwater + slope + liquefaction potential

EXPOSURE: people + buildings + lifelines + time of occurrence

VULNERABILITY: weak design + irregular form + poor maintenance

SYSTEM / DEFINITION

QUESTION: Why can similar earthquakes produce unequal disasters?

HAZARD: source size + depth + rupture geometry + path

SITE: basin sediment + groundwater + slope + liquefaction potential

EXPOSURE: people + buildings + lifelines + time of occurrence

PROCESS / MECHANISM

VULNERABILITY: weak design + irregular form + poor maintenance

CAPACITY: codes + enforcement + retrofit + warning + response

RISK VERDICT: hazard becomes disaster through social and spatial filters

15-MARK INDIA SPINE tectonic regions → site modifiers → exposure/vulnerability → cases

SPATIAL / INDIA PATTERN

zonation and microzonation → construction and governance → conclusion

TRAPS: focus is not epicentre

magnitude is not intensity tsunami is not a tide

caldera is not an ordinary crater zone map is not prediction

HAZARD / LIMIT / RESPONSE

code publication is not enforcement

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: zonation and microzonation → construction and governance → conclusion.

E EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

ADVANCED 1 — FINITE-FAULT RUPTURE, DIRECTIVITY AND FREQUENCY-DEPENDENT SITE

ADVANCED 2 — SEISMIC CYCLE, PALAEOSEISMOLOGY AND PROBABILITY

ADVANCED 3 — CODE HIERARCHY, PERFORMANCE AND RETROFIT TRIAGE

ADVANCED 4 — EARTHQUAKE EARLY WARNING AND OPERATIONAL LIMITS

ADVANCED 5 — VOLCANIC MONITORING AND ERUPTION-STYLE UNCERTAINTY

DIRECTIVITY HELPS EXPLAIN ELONGATED DAMAGE PATTERNS THAT A SIMPLE

FREQUENCY CONTENT MATTERS

OPTIONAL PROCESS DEPTH

ADVANCED 1 — FINITE-FAULT RUPTURE, DIRECTIVITY AND FREQUENCY-DEPENDENT SITE RESPONSE

ADVANCED 2 — SEISMIC CYCLE, PALAEOSEISMOLOGY AND PROBABILITY

ADVANCED 3 — CODE HIERARCHY, PERFORMANCE AND RETROFIT TRIAGE

ADVANCED 4 — EARTHQUAKE EARLY WARNING AND OPERATIONAL LIMITS

DATA / INSTITUTION LIMIT

ADVANCED 5 — VOLCANIC MONITORING AND ERUPTION-STYLE UNCERTAINTY

Directivity helps explain elongated damage patterns that a simple epicentral circle cannot.

Frequency content matters: short stiff structures respond differently from tall flexible structures.

Resonance occurs when dominant ground periods approach a structure's natural period.

CONTEMPORARY PARALLEL

Basin-edge effects can focus or convert waves; deep basins may prolong long-period shaking.

Nonlinearity: very strong shaking can change soil stiffness and damping, so weak-motion amplification cannot always be extrapolated directly.

earthquake swarms and volcanic tremor;

deformation from GPS, tilt and satellite radar;

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.